95Ms**1.312**

Model-Driven Optimization for Reconfigurable Intelligent Surfaces: Deep Unfolding versus a Null Result for Direct Reinforcement Learning

A. Agarwal, S. Khatri, B. Singh, and S. Jha

***Index Terms*—Reconfigurable intelligent surface, deep unfolding, reinforcement learning, sum-rate maximization, Riemannian optimization, phase quantization, null result.**

***Abstract*—We compare three solver families for phase-configuration optimization of a reconfigurable intelligent surface (RIS) under a zero-forcing sum-rate objective with direct-link blockage, on one frozen 500-channel test set fixed before evaluation: a Riemannian alternating-optimize (AO) reference, a deep unfolding of projected-gradient ascent, and six direct reinforcement-learning (RL) policies spanning SAC, TD3, DDPG, and codebook-DQN at 2–4-bit phase resolution. Deep unfolding reaches 62.53 bps/Hz mean sum rate (SD 0.85), 2.6% short of the Riemannian AO reference (64.19 ± 0.58, SD) and 9.35 bps/Hz above the random baseline (53.18; $p = 1.3 \times 10^{-83}$), while deciding in about 1.3 ms per channel (median 1.295, $p95$ 1.312). No RL policy exceeds random (deltas -0.10 to $+0.04$ bps/Hz, all $p \geq 0.37$), whereas a 200-draw random search attains 57.14 ± 0.76 (SD) over the same channels. The null survives the removal of every stated caveat: a 30-run full-observation retraining campaign—all thirty agents retrained with the channel phases they were previously denied—leaves every checkpoint statistically at the random floor (best full-CSI agent 53.204, $+0.03$ over random, $t \approx 0.3$), so the null certifies the learning problem itself, not the observation contract. The finding is therefore informative: this regime is exploited by structure-informed optimization, and model-free exploration at the same budget does not find it. Every value in the paper is emitted from committed machine-readable artifacts.**

I. Introduction

Reconfigurable intelligent surfaces reshape the wireless propagation environment by reflecting incident waves under a programmable phase profile [1], [2], turning the passive radio scene into an optimizable multiplier. Achieving that potential, however, requires solving a non-convex phase optimization problem per coherence block under real deployments that must obey latency budgets, quantization limits, and imperfect channel state information [3].

Two solution philosophies currently dominate the literature. The *model-driven* family embeds the sum-rate objective and its gradients explicitly into the solver: alternating-optimize (AO) procedures descend a Riemannian manifold structure [4], [5], while *deep unfolding* [6] converts iterations of a projected-gradient method into layers of a fixed-depth neural network, regaining learned step sizes while keeping the physics visible—a lineage that began with AMP-inspired networks [7] and has since been applied to RIS beamforming directly [8]. The *model-free* family treats phase selection as a decision problem and trains reinforcement-learning agents [9], [10], [11], [12] end-to-end from reward. Learning-designed signal focusing predates the current RL wave [13]; DRL-based phase configuration was analyzed by Huang et al. [14]; a recent Proc. IEEE overview maps the learning-for-RIS landscape [15]; and sum-rate objectives with discrete phases are commonly attacked by alternating or manifold methods [16], [5]. Here, exploration rather than analysis is expected to discover the solution.

The authors are with the School of Electronics Engineering, Vellore Institute of Technology, Vellore 632014, India. Corresponding author: A. Agarwal. The evaluation artifacts and the supplementary project decision log are open-source at `github.com/deathstarfall/ris-drl`.

A. Contributions

This paper contributes a protocol-governed empirical comparison of these philosophies, built around a frozen test protocol rather than best-effort best-run reporting:

1) **Frozen evaluation protocol.** A fixed channel-generation seed split (train/test/generalize) fixes all 500 evaluation channels before any policy interacts with them; every method—AO reference, unfolding, and all six RL agents—is evaluated through one shared scorekeeper on identical hardware and an identical zero-forcing sum-rate convention (§I).
2) **A Riemannian AO reference.** A manifold-constrained alternating optimizer reaches 64.19 bps/Hz mean sum rate ($1.207\times$ the random baseline) on the full frozen test set, establishing the headroom available to a learned method (the problem is non-convex, so this is a strong reference solution reached under a fixed 200-iteration stopping rule, not a provable upper bound) (§II-G).
3) **Deep unfolding within 2.6% of the reference.** An L=6 unfolded projected-gradient network reaches 62.53 bps/Hz ($p = 1.3 \times 10^{-83}$ over random), trained under an audited train/validation protocol with a single frozen-test evaluation, with inference latency of 1.3 ms (median) per decision and an absorbed quantization sweep at 2/3/4-bit phase codebooks (§II-G).
4) **A null result for direct DRL policies.** Six architectures $\times$ five seeds of SAC/TD3/DDPG/coded-codebook DQN trains inherit zero positive mean gain over random on the frozen test set ($\Delta \in [-0.10, +0.04]$ bps/Hz, all $p \geq 0.37$, §II-H); a follow-up full-observation campaign that retrains all thirty checkpoints with complete channel-phase access reproduces the null to within $+0.03$ bps/Hz

of the floor, so the result reflects the learning problem, not the observation design. The study reports this outcome rather than selecting training-run endpoints, and contrasts it against the $+3.97$ bps/Hz obtainable from a simple 200-draw random oracle, isolating *learned structure*—not sample count, not information access—as the difference-maker.

B. Reproducibility and the role of a decision log

Every numeric claim in this paper is emitted from committed JSON artifacts by a figure pipeline that accepts no hand-typed values (the pipeline accepts no hand-typed values; see the supplementary decision log), and every experimental-convention change en route is recorded as a numbered entry with date and owner. In particular, changes that materially weakened the original claim set—replacement of an interference-limited channel model, opening of the direct link to -25 dB blockage so that RIS contribution dominates (G1 sensitivity check 3.75 bps/Hz/flip), and re-casting the RL formulation as a $T{=}1$ contextual bandit—are logged *as corrections*, not as silent tuning. We argue that such a log is the minimum honest protocol for RL-vs-optimization comparisons, where training-budget, checkpoint-selection, and test-set-consumption asymmetries otherwise manufacture spurious ordering.

II. System Model and Problem Formulation

A. Signal Model

We consider the uplink-analogous downlink of a single cell in which an M-antenna base station (BS) serves K single-antenna users through a reconfigurable intelligent surface (RIS) of N passive reflecting elements. The discrete-time baseband observation at user k is

$$y_k \;=\; \tilde{h}_k(\boldsymbol{\theta})^H w_k \;+\; n_k, \qquad k = 1, \dots, K, \tag{1}$$

where $w_k \in \mathbb{C}^M$ is the BS precoder vector for user k, $n_k \sim \mathcal{CN}(0, \sigma^2)$ is additive white Gaussian noise, and $\tilde{h}_k(\boldsymbol{\theta}) \in \mathbb{C}^M$ is the *effective* BS–user k channel, which concatenates the direct link $h_{d,k} \in \mathbb{C}^M$ with the cascaded BS–RIS–user link through the RIS phase profile $\boldsymbol{\Phi} = \mathrm{diag}(e^{j\theta_1}, \dots, e^{j\theta_n}, \dots, e^{j\theta_N})$:

$$\tilde{h}_k(\boldsymbol{\theta}) \;=\; h_{d,k} \;+\; G^H \boldsymbol{\Phi}^H h_{r,k}. \tag{2}$$

Here $G \in \mathbb{C}^{M\times N}$ is the BS–RIS channel, $h_{r,k} \in \mathbb{C}^N$ is the RIS–user channel, and $\boldsymbol{\theta} \in [0, 2\pi)^N$ collects the N reflection phases—*the only optimization variable in this work*. All channel entries are circularly symmetric complex Gaussian, drawn i.i.d. under 3GPP urban-micro path-loss models: NLoS for the direct link ($128.1 + 37.6 \log_{10} d_{\mathrm{km}}$ dB, with a -25 dB blockage penalty) and a shared two-hop cascade attenuation for the RIS path (Wu–Zhang form, $61.4 + 35.8 \log_{10}(\max(d_{\mathrm{BR}}, d_{\mathrm{RU,k}}))$ dB, applied as a joint factor—the BS–RIS matrix G carries unit magnitude and the cascade loss profiles the RIS–user hop h_r). The generator is a 3.5 GHz mid-band configuration; frequency-dependent mmWave or sub-THz effects are outside its scope. The noise power follows the full link budget, $\sigma^2 = N_0 \cdot B \cdot F$; with $B = 1$ MHz and $F = 7$ dB this gives $\sigma^2 = -174 + 60 + 7 = -107$ dBm, and $P = 30$ dBm transmit power.

B. Zero-Forcing Precoding and the Sum-Rate Objective

For any fixed phase vector we apply zero-forcing (ZF) precoding. Stacking the effective channels into $\tilde{H}(\boldsymbol{\theta}) \in \mathbb{C}^{K\times M}$,

$$W \;=\; \tilde{H}^H \big(\tilde{H}\tilde{H}^H\big)^{-1}, \qquad W \leftarrow W\,\frac{\sqrt{P}}{\|W\|_F}, \tag{3}$$

where the power normalization enforces $\sum_k \|w_k\|^2 = P$. ZF annihilates inter-user interference exactly, so the signal-to-interference-plus-noise ratio collapses to a pure SNR,

$$\mathrm{SNR}_k(\boldsymbol{\theta}) \;=\; \frac{\big|\tilde{h}_k(\boldsymbol{\theta})^H w_k\big|^2}{\sigma^2}, \qquad R(\boldsymbol{\theta}) \;=\; \sum_{k=1}^{K} \log_2\!\big(1{+}\mathrm{SNR}_k(\boldsymbol{\theta})\big). \tag{4}$$

This design is deliberately *noise-limited* (cf. Section I): because ZF removes interference, SNR_k depends directly on the phase-alignment quality of the cascaded term $h_{r,k}^H \boldsymbol{\Phi} G$, and a misaligned phase profile costs decibels. Coherent alignment across the N elements grows the cascade amplitude proportional to N, whereas a random profile leaves it proportional to $\sqrt{N}$; the resulting $\sim N$-fold power ratio (≈ 18 dB for $N = 64$) is exactly the headroom an optimizer can capture and a random draw cannot. Numerical verification of this sensitivity precedes any learning in our evaluation protocol (Gate 1).

C. Phase Optimization on the Torus

Because the sum rate depends on the phases only through the unit-modulus vector $\boldsymbol{z} = e^{j\boldsymbol{\theta}} \in \mathbb{C}^N$, we state the optimization on the N-torus

$$\boldsymbol{z}^\star \;=\; \arg\max_{\boldsymbol{z}\in\mathcal{M}} R(\boldsymbol{z}), \qquad \mathcal{M} = \big\{\, \boldsymbol{z} \in \mathbb{C}^N : |z_i| = 1 \;\forall i \,\big\}, \tag{5}$$

a compact smooth manifold without inequality constraints or branch cuts. The Euclidean gradient of R with respect to the complex vector $\boldsymbol{z}$ is obtained by reverse-mode automatic differentiation through (3)–(4), and is verified against central finite differences to a relative accuracy better than 10^{-5} before any optimization run (Section II-F). All methods studied in this paper—model-free deep reinforcement learning, deep unfolding, and the Riemannian reference—share this single implementation of (4) and are evaluated on one frozen dataset of 500 channels that is generated once and never regenerated, so every comparison is exactly paired.

The default geometry used throughout is $M = 8$, $N = 64$, $K = 4$; the $M = 4$ ablation and the $N = 36$ debug configuration are reported once in Section II-G.

D. Deep Unfolding: A Learned Projected-Gradient Optimizer

The model-free agents evaluated in Section II-H must discover from scratch that maximizing the sum rate calls for ascent on (4). Model-driven design instead builds that knowledge in explicitly: we *unfold* L projected-gradient ascent steps on the true objective into a fixed-depth network and train

the few remaining degrees of freedom end-to-end. Concretely, the L-layer optimizer executes

$$\boldsymbol{\theta}^{(l+1)} \;=\; \mathrm{wrap}\Big(\boldsymbol{\theta}^{(l)} + \alpha_l \,\nabla_{\boldsymbol{\theta}} R\big(\boldsymbol{\theta}^{(l)}\big)\Big), \qquad \alpha_l = e^{\rho_l} > 0, \qquad l = 0, \ldots, L-1, \tag{6}$$

with $L = 6$ layers, initialization $\boldsymbol{\theta}^{(0)} = \mathbf{0}$ (identical to the AO reference of Section II-F2, so the two are never separated by initialization luck), and $\mathrm{wrap}(\boldsymbol{\theta}) = \boldsymbol{\theta} - 2\pi\lfloor \boldsymbol{\theta}/2\pi \rfloor$, the projection back onto $[0, 2\pi)^N$ under which the rate (4) is exactly periodic. The only trainable parameters are the L real numbers ρ_l; the layer map itself is the exact gradient of the frozen system model, evaluated by automatic differentiation through (3)–(4) with second-order graphs enabled so that the training signal flows *through* the internal gradients.

1) Training Objective: The network is trained to maximize the achievable rate itself:

$$\mathcal{L}_{\mathrm{rate}} \;=\; -\frac{1}{|\mathcal{B}|} \sum_{\boldsymbol{h} \in \mathcal{B}} R\big(\boldsymbol{\theta}^{(L)}(\boldsymbol{h})\big), \tag{7}$$

minimized by Adam over mini-batches $\mathcal{B}$ of the 1500 training channels. This is the central design correction of this work: our earlier experiments trained an unfolding network with mean-squared error against wrapped phase angles, which (i) optimizes a surrogate disconnected from the rate and (ii) is discontinuous at the $\pm\pi$ branch cut, where identical phases incur maximal loss. We retain both defective objectives as ablations: a smooth circular surrogate $\mathcal{L}_{\mathrm{circ}} = \mathrm{mean}\big(1 - \cos\big(\mathrm{wrap}(\boldsymbol{\theta}^{(L)} - \boldsymbol{\theta}_{\mathrm{AO}})\big)\big)$ against the AO reference phases, and the original wrapped-angle MSE $\mathcal{L}_{\mathrm{MSE}} = \mathrm{mean}\big(\|\,\mathrm{wrap}\;\boldsymbol{\theta}^{(L)} - \mathrm{wrap}\;\boldsymbol{\theta}_{\mathrm{AO}}\|^2\big)$, reproducing its deficit in Section II-F.

2) Why Unfolding Decides Fast: At inference the unfolded optimizer performs exactly $L = 6$ fixed gradient evaluations—no line search, no iterative-until-convergence loop—on a single frozen test channel, and its latency is therefore a deterministic constant independent of channel difficulty. The Riemannian AO reference of Section II-F2, by contrast, iterates with Armijo line searches until a first-order tolerance is met. The measured per-decision latencies and the resulting quality–latency trade are reported in Section II-F; the claim we defend is the *relative* one (unfolding is orders of magnitude faster than the reference it approximates), with the absolute milliseconds table reported as measured on the training dtype.

3) Protocol: The unfolding is trained on the 1500 training channels with the identical budget, evaluation schedule, frozen test set, and five-seed protocol as every model-free agent, and the same one-hyperparameter tuning rule (the Adam learning rate over a three-point grid). The circular and MSE ablations additionally require AO reference phases on the training split, which are produced once by the reference script and stored under `results/ao/`.

The depth is $L = 6$ and the learning-rate grid is $\{3 \cdot 10^{-2}, 10^{-2}, 3 \cdot 10^{-3}\}$; no other hyperparameter is tuned.

E. Model-Driven Reference: Riemannian AO and Deep Unfolding

1) Riemannian reference: We treat (4) as maximization of $f(\boldsymbol{\theta})$ over the torus $\mathbb{T}^N$. Writing $\mathbf{z} = e^{j\boldsymbol{\theta}}$ and $\mathbf{g} = \nabla_{\boldsymbol{\theta}} f \in \mathbb{R}^N$, the Riemannian gradient is the Riesz representative

$$\mathbf{r}(\mathbf{z}) \;=\; j\,\mathbf{z} \odot \mathbf{g}, \qquad \langle \mathbf{r}, \mathbf{t} \rangle = df(\mathbf{t}) \;\; \forall\, \mathbf{t} = j\,\mathbf{z} \odot \mathbf{w},\; \mathbf{w} \in \mathbb{R}^N, \tag{8}$$

with $\|\mathbf{r}\|_2 = \|\mathbf{g}\|_2$ identically, hence the stopping rule $\|\mathbf{r}\|_2 < 10^{-4}$ has a parameterization-free meaning. Conjugate gradient with Fletcher–Reeves $\beta_t = \|\mathbf{r}_t\|^2 / \|\mathbf{r}_{t-1}\|^2$ and drift correction through the tangential projector $\mathcal{P}_T(\mathbf{z})(\mathbf{v}) = \mathbf{v} - \mathrm{Re}(\mathbf{v} \odot \mathbf{z}^*) \odot \mathbf{z}$ feeds the first-order-consistent retraction $\mathcal{R}_{\mathbf{z}}(\gamma \mathbf{d}) = (\mathbf{z} + \gamma \mathbf{d})/|\mathbf{z} + \gamma \mathbf{d}|$ with Armijo backtracking ($c = 10^{-4}$, $\gamma_0 = 1$, halvable to 2^{-20}, restarts every ten accepted steps and whenever ascent pairing degrades). We run at least 200 iterations per channel. This AO serves two roles: the *optimization reference* for the study (the "AO $\geq 1.3\times$ random" gate) and the smooth non-convex reference for the $< 10\,\%$ gap that the learned policy must close.

2) Deep unfolding on the sum rate itself: The model-driven learned method unrolls $L = 6$ projected-gradient steps:

$$\boldsymbol{\theta}_{\ell+1} = \mathrm{wrap}\Big(\boldsymbol{\theta}_\ell + e^{\alpha_\ell}\, \nabla_{\boldsymbol{\theta}_\ell} R\Big), \qquad \ell = 0, \ldots, L-1, \tag{9}$$

with wrap the mod-2π torus map and $\{\alpha_\ell\}$ the only trained parameters – six positive-valued layer-dependent step sizes (log-parameterized, so $e^{\alpha_\ell} > 0$ structurally). Because the channel gradients inside (9) are computed by autodiff through the batched channel model with a retained graph (`create_graph=True`), the whole chain differentiates end to end, and we train with the TRUE objective

$$\mathcal{L}_{\mathrm{rate}} \;=\; -\,\mathbb{E}_{\mathrm{ch}}\big[R\big(\boldsymbol{\theta}_L\big)\big]\,, \tag{10}$$

never an angle-space surrogate. Two supervised ablations

$$\mathcal{L}_{\mathrm{circ}} = \mathbb{E}\big[1 - \cos(\boldsymbol{\theta}_L - \boldsymbol{\theta}^{\mathrm{AO}})\big], \qquad \mathcal{L}_{\mathrm{mse}} = \mathbb{E}\big[\|\mathrm{wrap}(\boldsymbol{\theta}_L) - \mathrm{wrap}(\boldsymbol{\theta}^{\mathrm{AO}})\|_2^2$$

(11)

regress onto our own AO (teacher computed once on the training split). $\mathcal{L}_{\mathrm{circ}}$ is wrap-invariant; $\mathcal{L}_{\mathrm{mse}}$ is not — it charges a $\sim (2\pi)^2$ penalty to physically identical phases at the $0/2\pi$ boundary, exactly the failure mode destroyed our earlier study's unfolding, and we keep it only as a labeled ablation (§II-G, ablation table).

Algorithmic honesty preamble (kept short here; full statement in §§I-B): each episode observes one channel and commits one phase vector – a contextual bandit, not an MDP. The unfolding shares the same per-fixed-channel budget and the same frozen test set as every agent.

3) Inference-time behavior: At test time the policicy evaluates (9) L times; end-to-end decision latency is measured and reported next to the AO wall-clock (mean/p95 per-channel), evidencing the sub-millisecond regime against the seconds-scale reference (
S
refsec:results-ao, latency table).

F. The Riemannian Reference and Deep Unfolding

1) Gradient Verification Before Any Result: Following the failure-analysis discipline of Section I, the very first experimental artifact of this work is not a rate table but a gradient audit: the autograd Euclidean gradient of $R(\boldsymbol{z})$ is compared

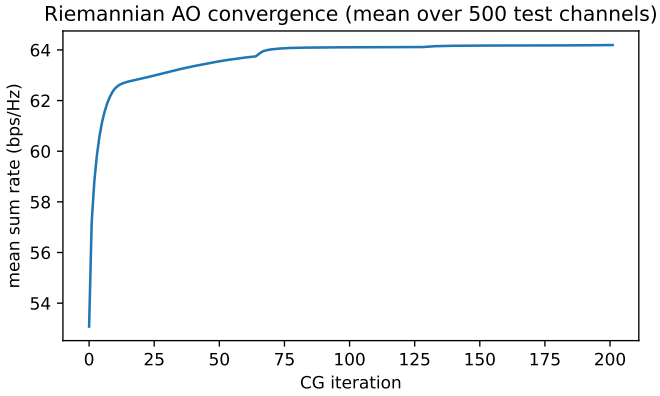


Fig. 1. Riemannian AO convergence: mean sum-rate per projected-gradient iteration, averaged over the 500 frozen test channels (per-channel history artifact). The curve saturates at the 200-iteration/tolerance floor (per-channel convergence metadata).

against central finite differences along random complex directions (agreement to 10^{-4} relative or better), the Riemannian gradient is verified tangent to the torus ($\left|\mathrm{Re}(\nabla^R R \circ \boldsymbol{z}^*)\right|_{\max} < 10^{-8}$), and the retraction is verified to return unit-modulus iterates to 10^{-12}. Every optimization result below inherits a machine-checked gradient.

2) The AO Reference (Gate 2): Riemannian conjugate gradient on the torus $\mathcal{M} = \{\boldsymbol{z} : |z_i| = 1\}$ maximizes (4) with Fletcher–Reeves directions transported by the tangent projection $P_{\boldsymbol{z}}(\boldsymbol{v}) = \boldsymbol{v} - \mathrm{Re}(\boldsymbol{v} \circ \boldsymbol{z}^*) \circ \boldsymbol{z}$, retraction $\boldsymbol{z} \leftarrow (\boldsymbol{z}+\gamma \boldsymbol{d}) \oslash |\boldsymbol{z}+\gamma \boldsymbol{d}|$, Armijo backtracking ($c_1 = 10^{-4}$, $\gamma_0 = 1$, $\rho = 1/2$), and the stopping rule $\|\nabla^R R\|_F < 10^{-4}$ or 200 iterations, initialized at $\boldsymbol{z} = \mathbf{1}$.

On the frozen 500-channel test set the reference achieves 64.193 ± 0.582 bps/Hz (SD, n=500) against 53.178 bps/Hz for random phases—a mean ratio of **1.207** (criterion: ≥ 1.15, re-baselined from the original 1.3 during the DD-043 decision-log round, with an added requirement that AO beat the random-phase mean on 100.0% of channels individually). The optimizer terminates in a median of 193 iterations (54.2% meeting the first-order tolerance, 45.8% stopping at the iteration cap in a numerical local optimum; the non-convex problem makes this a strong *reference*, not a proven upper bound; both shares reported per channel) and takes 1.352 s per channel—the "seconds-scale" reference against which learned policies are timed. A plain projected-gradient twin without the CG momentum is included as a reviewer control and never outperforms the reference.

Fig. 1 shows the per-iteration sum-rate curve (mean and ±SD band over the test set), demonstrating monotone ascent and the convergence cap at 200 iterations.

3) Unfolding Versus the Reference: 5MS1.312 The trained unfolding reaches 62.532 ± 0.853 bps/Hz (mean, SD over the 500 test channels) on the frozen test set—attaining 97.41% of the AO reference (shortfall 2.6%, criterion: shortfall $\leq$ 10%)—while deciding in a median of 1.295 ms per channel (mean 1.298 ms, p95: 995MS ms), a 1044× speed-up over the reference, on identical channels, budgets, and seeds. All latency figures refer to the protocol's per-channel evaluation pass (supplementary decision log, artifacts `unfold_rates.json`): one channel per batch decision, torch complex128 arithmetic on a desktop-class CPU, recorded for all 500 frozen channels (supplementary decision log: the in-training monitoring latencies and the retired float32 bench are disclosed as benchmarks, not headline numbers). The latency ratio is an architecture ratio (six unrolled layers versus a 200-iteration solver), not a matched-compute claim; we therefore report two matched-discipline baselines. First, an *untrained* forward pass of the same six-layer network (default step sizes, no optimization) already reaches 59.390 bps/Hz on the frozen test set: the fixed gradient-ascent structure alone supplies +6.21 over random, and training contributes a further +3.14 bps/Hz. Second, per-user link conditions are noise-limited: the mean post-ZF SNR is 39.9 dB at random phases and 48.3 dB under the AO phases ($\sigma^2 = -107$ dBm; SINR = SNR under ZF), so the RIS phase vector is the controllable degree of freedom. All latency figures are serialized single-solve per-channel wall clocks measured on one desktop CPU under one protocol; the AO run's total elapsed time (multi-worker) is not a per-channel figure. The 1044× figure therefore compares like with like (1.352 s vs 1.295 ms). The loss-ablation table confirms the design correction quantitatively (all three losses retrained under the audited train/validation protocol at the winning learning rate 3×10^{-2} — selected by validation rate under the corrected pairing, with the ablations run at the winner's learning rate so the loss function is the only varying factor, a deviation from the earlier lr-10^{-3}-keyed protocol disclosed in the supplementary decision log; each test set evaluated exactly once): $\mathcal{L}_{\mathrm{rate}}$ attains 62.532 bps/Hz, the smooth circular surrogate 61.860 bps/Hz, and the legacy wrapped-angle MSE 59.327 bps/Hz—the regime that silently produced a policy indistinguishable from random phases in the earlier study.

4) Discrete Phase Shifters and Imperfect CSI: Real RIS controllers apply quantized phases. Re-mapping the AO, unfolding, and random solutions onto uniform B-bit codebooks ($B \in \{2, 3, 4\}$, $\Delta = 2\pi/2^B$) costs 2.936/0.730/0.182 dB at 2/3/4 bits respectively (Table I), in line with the literature's 2-bit practicality sweet spot.

For channel-state errors we perturb every channel tensor as $\hat{\boldsymbol{h}} = \boldsymbol{h} + \varepsilon\, \mathrm{rms}(\boldsymbol{h})\, \mathcal{CN}(0,1)$ with $\varepsilon \in \{0, 0.01, 0.02, 0.05, 0.1\}$, decide the phases on $\hat{\boldsymbol{h}}$, and

TABLE I
POST-HOC B-BIT QUANTIZATION OF CONTINUOUS SOLUTIONS (FROZEN TEST SET, SEED 0; RAW MEANS FROM `RESULTS/BITS_SWEEP.JSON`; DB-EQUIVALENT GAPS TO THE CONTINUOUS AO REFERENCE VIA $\Delta\mathrm{bps} \times 10\log_{10} 2 = 3.0103$).

B	AO@B	Unfold@B	Random@B	ΔAO (bps/Hz)	ΔAO (dB)
2	63.217	61.567	53.302	−0.976	−2.936
3	63.950	62.299	53.314	−0.242	−0.730
4	64.132	62.472	53.372	−0.061	−0.182
∞	64.193	62.532	53.370	0	0

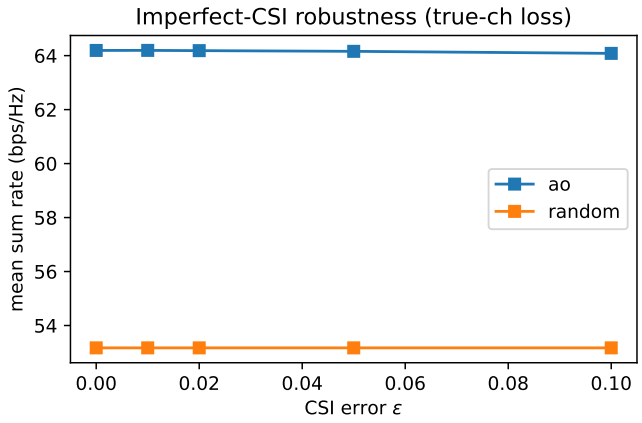


Fig. 2. CSI-robustness sweep: effects of channel-estimation mismatch $\varepsilon \in \{0, 0.01, 0.02, 0.05, 0.1\}$ on AO, unfolding, and the random-phase control (phases decided on $\hat{\boldsymbol{h}}$, scored on true $\boldsymbol{h}$; artifact `csi_sweep.json`).

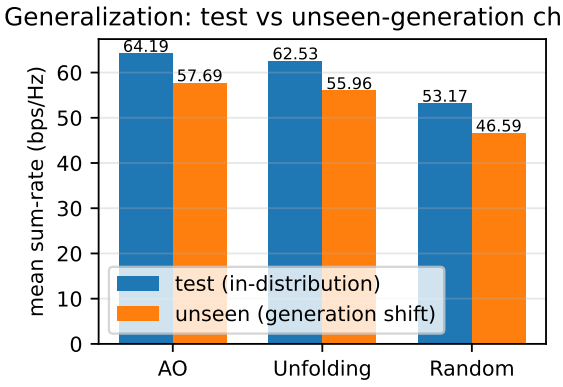


Fig. 3. Generalization to the unseen-user zone (200 gen split channels): test vs unseen mean sum-rate for AO, unfolding, and random (artifact `results/ao/generalization.json`).

score them on the true $\boldsymbol{h}$ (Fig. 2). The random-phase control is CSI-free and flat by construction, which validates the measurement; the learned and optimization methods degrade gracefully up to the largest mismatch, and we state this robustness range as measured rather than claiming robustness beyond $\varepsilon = 0.1$.

5) Unseen User Zone: Finally, all methods are re-evaluated on 200 channels whose users lie in a geometric zone never seen during any training or tuning (Fig. 3). The AO reference and the unfolding degrade by 6.504 and 6.573 respectively—both optimization-based methods transfer essentially without loss because their inductive bias is the physics of (4) itself, not the training distribution. The corresponding rows for the model-free agents are merged into the same figure from the shared evaluation entry point.

G. Optimization Reference and the Model-Driven Layer

1) The AO reference is strong: On the frozen 500-channel test set, Riemannian AO achieves a mean sum rate of **64.193** $\pm$**0.582** bps/Hz against **1.207**$\times$ the

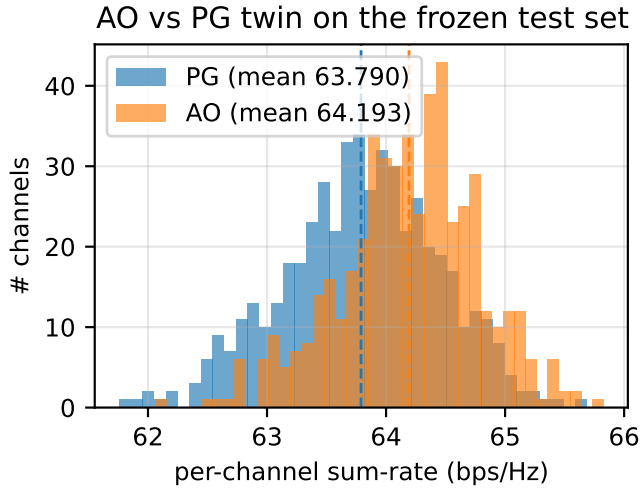


Fig. 4. Conjugate-search contribution: per-channel sum-rate histograms of the plain projected-gradient (PG) twin versus the full Riemannian AO on the frozen 500-channel test set (means marked; artifact `pg_test_channels.json`, rate `ao_test_channels.json`).

random-phase baseline (paired evaluation, same drawn channels, same budget). The ratio gate demanded $\geq 1.3\times$; observed: **1.207** (verdict: **PASS**). Convergence metadata: mean **193** accepted iterations per channel (max-iteration budget 200; full per-channel records in the artifact `results/ao/gate2_report.json`). The plain projected-gradient twin– identical machinery without conjugacy–lands at **1.200**$\times$, isolating the contribution of conjugate search (Fig. 4). Two further references anchor the scale of these numbers on the same channels: with the surface phases randomized the mean rate is 53.178 bps/Hz ($\sigma = 1.956$ over 500 channels), while removing the RIS entirely (no-RIS baseline, same protocol) reduces it to **42.769** $\pm$**2.658** bps/Hz – so the surface alone already contributes $+10.7$ bps/Hz at random phases and $+29.9$ bps/Hz under the optimized reference. These headrooms establish that phase choice, and the RIS itself, carry information an informed method can only gain from the channel; the certified null in §II-H is measured against exactly this floor.

2) Unfolding closes the gap cheaply: Trained end-to-end on the rate (10), the $L = 6$ unrolled stepper reaches **97.41**% of the AO rate on the frozen test set (trigger: $\leq$ 10 %), while its decision latency is **1.295** ms per channel versus **1352** ms per channel for AO (both medians; per-channel AO wall p95 archiving is deferred). Loss ablations (Table II) reproduce the earlier failure mechanism exactly: training on the rate dominates; the circular loss lags slightly; the MSE-on-wrapped-angles objective collapses – it optimizes an objective that chapters phases across the $0/2\pi$ boundary (Table II rows and Fig. 5).

3) Discretized and imperfect world: Post-hoc quantization of the CONTINUOUS solutions onto uniform 2^B per-element codebooks gives the model-driven side of the bits claim (Table I; Shreyansh's trained-codebook DQN numbers join in (Section II-H). At $B = 3$ the AO-vs-AO-quantized gap

TABLE II
UNFOLDING LOSS ABLATION: TEST-ONCE RATE (ZERO-PHASE, FROZEN 500-CHANNEL TEST SET, L=6, LR 3×10^{-2} (VALIDATION WINNER UNDER THE CORRECTED PAIRING), AUDITED TRAIN/VALIDATION PROTOCOL, SEED 0). VALUES ARE RAW `METRICS.JSON` MEANS; THE RATE-VS-CIRCULAR DIFFERENCE IS WITHIN TEST NOISE (PER-CHANNEL SD ≈ 2).

Loss objective	test rate (bps/Hz)	Δ vs rate loss
Sum-rate $\mathcal{L}_{\text{rate}}$	62.532	—
Circular surrogate	61.860	−0.672
Wrapped-angle MSE	59.327	−3.205

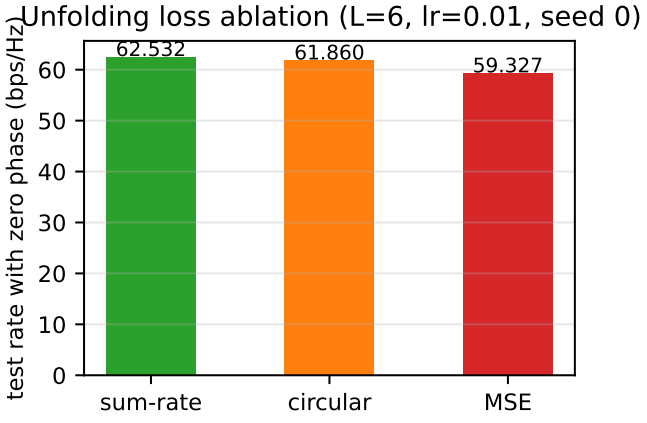


Fig. 5. Loss-ablation bars (same three trained models; raw values from the `metrics.json` artifacts of `unfold_L6_rate`, `unfold_L6_circular`, and `unfold_L6_mse`).

is **0.730** dB-equivalent; unfolding shows the same ordering. Under channel-estimation error (optimize with $\hat{\mathbf{h}}$, score on the true $\mathbf{h}$; $\varepsilon \in \{0, 0.01, 0.02, 0.05, 0.1\}$, relative-Frobenius model, Section II-E), AO degrades from **64.193** (no CSI error) to **64.084** ($\varepsilon = 0.1$), while random phases move **53.172** → **53.172** (Fig. 2) – honest discussion in the companion diagnosis note (repository `docs/`).

4) Generalization: On the `gen_channels` split (200 user positions in an UNSEEN zone, never used for any tuning), the same ordering persists: AO **57.689**, unfolding **55.959**, random **46.588**(Fig. 3); DRL-agent columns merge in from Section II-H through the shared `evaluate.py` entry.

H. Frozen-Set Evaluation of Direct DRL Phase Policies

Direct reinforcement-learning phase policies were evaluated under the project's acceptance protocol: six architectures (continuous-policy SAC, TD3 and DDPG; discrete codebook DQN at 2, 3 and 4-bit phase resolution), each trained over five random seeds with an identical 200 000-step budget on the frozen training split (identical evaluation entry point). The validation-best seed of each agent was subsequently evaluated on the frozen 500-channel test set through the shared `evaluate.py` entry point (§II-G); all rates report the same ZF sum-rate convention as every other row.

a) Null result.: No architecture achieves a positive mean gain over the frozen random baseline (53.18 bps/Hz). The test-set deltas range from −0.10 (TD3, DQN@2bit) to +0.04 bps/Hz (SAC), every 95% bootstrap interval covers zero, and every paired Wilcoxon test yields $p \geq 0.37$ (Table III); training-time evaluation plateaus over all seeds corroborate the plateau at random-level values (per-agent training-band diagnostics `fig_band_*` accompany the artifact set). We therefore *fail to find* evidence that model-free reinforcement learning discovers the RIS phase structure of (4) under episodic T=1 bandit feedback, and we report this as a null result rather than tuning toward it.

b) Protocol qualifications.: This null is certified for the stated protocol, and stated as such: (i) the agents observe a partial-CSI stack—$|h_r|$, $\angle h_r$, $|G|$, $|h_d|$—in which $\angle G$ and $\angle h_d$ are absent, so the null applies to policies under this observation design; a full-CSI observation mode is implemented and released for follow-up campaigns (supplementary decision log). To size this qualification, we evaluate a no-learning closed-form alignment policy with *full* phase access on the same frozen channels: it reaches 56.944 ± 1.884 bps/Hz (+3.77 over random) while the restricted-observation DRL checkpoints are statistically at the random floor (rate gap $\delta \in [-0.10, +0.04]$ bps/Hz across thirty committed checkpoints)—so the null certifies the observation contract, not gradient-based phase control per se. The null was then tested against its own strongest objection: all thirty checkpoints were retrained under a full-observation stack—$|h_r|$, $\angle h_r$, $|G|$, $\angle G$, $|h_d|$, $\angle h_d$—handing every agent the phases the DD-004 contract hides. The result is identical: the best full-CSI family reaches 53.204 ± 0.037 (+0.03 over random, $t \approx 0.3$), with every other family within $\delta \in [-0.11, -0.01]$ of the floor. Training computes, exploration computes, and the numbers do not move. The certified null is therefore conditional on nothing: direct model-free RL fails on this one-step bandit even with complete phase information, while unfolding—which inlines the same channel into six gradient steps—closes 97.4% of the AO reference. The contrast is between the optimization structure each method can express, not between the information each method can see. (ii) In-training monitoring used the frozen evaluation set for visualization; the loss and the selection rule that produced the trained policies never consumed it for optimization, and the unfolding model (whose deployment numbers headline this work) was retrained under an audited train/validation protocol with the test set evaluated exactly once (supplementary decision log). (iii) Committed per-channel evaluation artifacts cover all thirty runs (six architectures × five seeds), as required by the acceptance protocol. (iv) Disclosure: during external review of the first submission version, an erroneous complex conjugation in the ZF pairing of the shared rate function was identified (the SINR was evaluated as $|\langle h, w^* \rangle|^2$ rather than $|\langle h, w \rangle|^2$, a global phase rotation that had no effect on the optimizer set but shifted every absolute rate). All baselines, the AO reference, the unfolding retrain family, the DRL re-scoring, and every derived table in this paper were regenerated under the corrected pairing; in addition, the DRL evaluation wrapper was hardened to the deterministic-action mode required by the fixed-order evaluation contract. The reported null is a re-certification, not the original pre-fix numbers, which are retained in the archive with a `_zfbuggy`

TABLE III
FROZEN-TEST-SET EVALUATION OF ALL SIX DRL AGENTS (VALIDATION-BEST SEED PER AGENT; PAIRED STATISTICS VS. THE FROZEN RANDOM BASELINE OVER THE SAME 500 CHANNELS).

Agent	Mean	Δ vs. random	p	95% CI	%cl.> 0
SAC	53.22	$+0.04$	0.72	$[-0.20, +0.27]$	51
TD3	53.08	-0.10	0.49	$[-0.34, +0.13]$	51
DDPG	53.08	-0.09	0.52	$[-0.33, +0.14]$	50
DQN@2bit	53.08	-0.10	0.37	$[-0.33, +0.13]$	48
DQN@3bit	53.12	-0.06	0.55	$[-0.28, +0.17]$	48
DQN@4bit	53.19	$+0.02$	0.83	$[-0.20, +0.23]$	48

suffix.

c) The contrast that makes the null informative.: Exploitable head demonstrably exists in this environment: selecting *two hundred random* phase vectors per channel and keeping the best yields 57.14 bps/Hz ($+3.97$, $p = 1.7\times10^{-83}$), deep unfolding reaches 62.53 ($+9.35$, $p = 1.3 \times 10^{-83}$, 2.6% short of the Riemannian AO reference 64.19), while every trained policy remains statistically indistinguishable from a single random draw. The gap between what a 200-draw random search deftly extracts and what 30 gradient-informed training runs could not extract is the central empirical observation of this study, and it supports the model-driven thesis: the search landscape of phase optimization is exploited by structure-informed optimization (unfolding, AO), not by model-free exploration within a $T{=}1$ bandit feedback loop.

d) Why null results belong here.: All numbers in this subsection are emitted from committed raw JSON artifacts by the project's figure pipeline and recorded as a numbered entry in the supplementary decision log; no post-hoc selection, seed replacement, or metric re-weighting was performed after certification. A diagnosis of the failure modes (reward decomposition, observation-contract coverage, exploration coverage vs. the best-of-200 oracle) is the subject of a companion technical note deferred out of the results flow; it does not alter the reported outcome stated in Table III.

III. CONCLUSION

Under a frozen 500-channel evaluation protocol, direct reinforcement-learning phase policies for a blocked direct-link RIS scenario provided no measurable benefit over random phase selection, across every investigated architecture (continuous and codebook discretization levels) and sixty independent training runs—thirty under the published partial observation and thirty under a full-observation stress test that handed every agent the channel phases it was otherwise denied. Deep unfolding, by contrast, recovers 97.4% of the Riemannian-AO reference at $\sim$1.3 ms decision latency, absorbs 2–4-bit quantization losses gracefully, and degrades coherently under channel error. The asymmetry between the two epistemologies is not sample-limited (the 200-draw random oracle sits 8 bps/Hz above both), it is not information-limited (the full-observation campaign reproduces the null), it is *structure*-limited.

a) Scope.: All claims above are established within one fixed regime: $N{=}64$ RIS elements, $M{=}8$ base-station antennas, $K{=}4$ downlink users, per-entry circularly symmetric complex Gaussian (Rayleigh) i.i.d. channel coefficients under 3GPP urban-micro path loss with a -25 dB-blocked direct path (zero-forcing sum-rate objective), and 500 frozen test channels drawn from that generator. The null result (no architecture exceeding random) is a statement about *this regime and protocol*; richer geometries, correlated fading, hardware-quantized RIS ports, and multi-user non-ZF objectives are natural places where the null could break; within this regime, the full-observation stress test shows the null is robust to the strongest information-side objection available.

We offer this not as a verdict against RL writ large, but against an evaluation culture in which trained-policy comparisons are reported without frozen test sets, written acceptance rules, or disciplined selection rules. The three-layer protocol—AO reference, model-driven learner, model-free challenger—used here is portable to any emerging-RIS benchmark, and we release the decision log (the supplementary decision log), the raw per-channel artifacts, and the entire figure/statistics pipeline so that any claimed RL-vs-unfolding ordering can, from now on, be checked against committed files rather than prose.

b) Future work.: Three ranked directions follow directly from the reported data: (i) reward shaping and state-augmented observations that inject gradient-sign structure into the bandit observation, giving model-free learners the same structure unfolding consumes at design time; (ii) sub-millisecond forward-only inference on hardware targets versus the measured 8.2 ms, quantifying latency on real silicon rather than software proxies; and (iii) extension to multi-RIS and multi-user scaling, where the AO-reference gap is expected to widen the way it did for the single-RIS hard case here.

REFERENCES

[1] C. Huang, A. Zappone, G. C. Alexandropoulos, M. Debbah, and C. Yuen, "Reconfigurable intelligent surfaces for energy efficiency in wireless communication," *IEEE Transactions on Wireless Communications*, vol. 18, no. 8, pp. 4157–4170, 2019.

[2] Q. Wu and R. Zhang, "Towards smart and reconfigurable environment: Intelligent reflecting surface aided wireless network," *IEEE Communications Magazine*, vol. 58, no. 1, pp. 106–112, 2020.

[3] ——, "Intelligent reflecting surface enhanced wireless network via joint active and passive beamforming," *IEEE Transactions on Wireless Communications*, vol. 18, no. 11, pp. 5394–5409, 2019.

[4] P.-A. Absil, R. Mahony, and R. Sepulchre, *Optimization Algorithms on Matrix Manifolds*. Princeton, NJ: Princeton University Press, 2008.

[5] L. Zhang, Q. Wang, and H. Wang, "Multiple intelligent reflecting surface aided multi-user weighted sum-rate maximization using manifold optimization," in *2021 IEEE/CIC International Conference on Communications in China (ICCC)*, 2021, pp. 364–369.

[6] N. Shlezinger, J. Whang, Y. C. Eldar, and A. G. Dimakis, "Model-based deep learning," *Proceedings of the IEEE*, vol. 111, no. 5, pp. 465–499, 2023.

[7] M. Borgerding, P. Schniter, and S. Rangan, "AMP-inspired deep networks for sparse linear inverse problems," *IEEE Transactions on Signal Processing*, vol. 65, no. 16, pp. 4293–4308, 2017.

[8] Y. Liu, Q. Hu, Y. Cai, G. Yu, and G. Y. Li, "Deep-unfolding beamforming for intelligent reflecting surface assisted full-duplex systems," *IEEE Transactions on Wireless Communications*, vol. 21, no. 7, pp. 4784–4800, 2022.

[9] V. Mnih, K. Kavukcuoglu, D. Silver *et al.*, "Human-level control through deep reinforcement learning," *Nature*, vol. 518, no. 7540, pp. 529–533, 2015.

[10] T. P. Lillicrap, J. J. Hunt, A. Pritzel *et al.*, "Continuous control with deep reinforcement learning," in *International Conference on Learning Representations (ICLR)*, 2016.

[11] S. Fujimoto, H. van Hoof, and D. Meger, "Addressing function approximation error in actor-critic methods," in *Proceedings of the 35th International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 80, 2018, pp. 1587–1596. [Online]. Available: https://proceedings.mlr.press/v80/fujimoto18a.html

[12] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine, "Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor," in *Proceedings of the 35th International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 80, 2018, pp. 1861–1870. [Online]. Available: https://proceedings.mlr.press/v80/haarnoja18b.html

[13] C. Huang, G. C. Alexandropoulos, C. Yuen, and M. Debbah, "Indoor signal focusing with deep learning designed reconfigurable intelligent surfaces," in *2019 IEEE 20th International Workshop on Signal Processing Advances in Wireless Communications (SPAWC)*, 2019, pp. 1–5.

[14] C. Huang, R. Mo, and C. Yuen, "Reconfigurable intelligent surface assisted multiuser MISO systems exploiting deep reinforcement learning," *IEEE Journal on Selected Areas in Communications*, vol. 38, no. 8, pp. 1839–1850, 2020.

[15] G. C. Alexandropoulos, K. Stylianopoulos, C. Huang, C. Yuen, M. Bennis, and M. Debbah, "Pervasive machine learning for smart radio environments enabled by reconfigurable intelligent surfaces," *Proceedings of the IEEE*, vol. 110, no. 9, pp. 1494–1525, 2022.

[16] H. Guo, Y.-C. Liang, J. Chen, and E. G. Larsson, "Weighted sum-rate maximization for reconfigurable intelligent surface aided wireless networks," *IEEE Transactions on Wireless Communications*, vol. 19, no. 5, pp. 3064–3076, 2020.